Practice Test · Class X · Mathematics

Trigonometry — Unit Test

Chapters: Introduction to Trigonometry · Some Applications of Trigonometry

Time: 1 hr 30 min **Maximum Marks: 40**

General Instructions: All questions are compulsory. Section A: 8 × 1 = 8 marks · Section B: 4 × 2 = 8 marks · Section C: 4 × 3 = 12 marks · Section D: 3 × 4 = 12 marks. Draw figures wherever necessary. Show all steps.

SECTION A

Multiple choice questions · 1 mark each

1. If $\sin A = 3/5$, then the value of $\cos A$ is: [1]

(a) 4/5 (b) 3/4 (c) 5/4 (d) 4/3

2. The value of $\sin 30° \cos 60° + \cos 30° \sin 60°$ is: [1]

(a) 0 (b) 1/2 (c) 1 (d) √3/2

3. $(1 + \tan^2 A)$ is equal to: [1]

(a) $\sec^2 A$ (b) $\operatorname{cosec}^2 A$ (c) $\cot^2 A$ (d) $\sin^2 A$

4. If $\tan \theta = 1$ and θ is acute, then θ is equal to: [1]

(a) 30° (b) 45° (c) 60° (d) 90°

5. The value of $9 \sec^2 A - 9 \tan^2 A$ is: [1]

(a) 1 (b) 9 (c) 8 (d) 0

6. The value of $\tan 60° \div \cot 30°$ is: [1]

(a) 0 (b) 1 (c) √3 (d) 1/√3

7. The angle of elevation of the top of a 30 m high tower from a point on the ground is 30°. The distance of the point from the foot of the tower is: [1]

(a) 30 m (b) 10√3 m (c) 30√3 m (d) 15 m

8. If $\cos A = 1/2$ and A is acute, then $\tan A$ is equal to: [1]

(a) 1/√3 (b) 1 (c) √3 (d) 2

SECTION B

Very short answer questions · 2 marks each

9. If $3 \cot A = 4$, find the value of $(1 - \tan^2 A) / (1 + \tan^2 A)$. [2]

10. Evaluate: $2 \tan^2 45° + \cos^2 30° - \sin^2 60°$ [2]

11. If $\sin (A - B) = 1/2$ and $\cos (A + B) = 1/2$, where $0° < A + B \leq 90°$ and $A > B$, find A and B. [2]

12. Prove that: $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2 = 2$ [2]

SECTION C

Short answer questions · 3 marks each

13. Prove that: $(\sec A - \tan A)^2 = (1 - \sin A) / (1 + \sin A)$ [3]

14. If $\sec \theta = 13/12$, find the values of all the other trigonometric ratios of θ. [3]

15. Prove that: $\cos A / (1 + \sin A) + (1 + \sin A) / \cos A = 2 \sec A$ [3]

16. A kite is flying at a height of 60 m above the ground. The string attached to the kite is tied to a point on the ground and makes an angle of 60° with the ground. Assuming there is no slack in the string, find the length of the string. [3]

SECTION D

Long answer questions · 4 marks each

17. Prove that: $\tan\theta / (1 - \cot\theta) + \cot\theta / (1 - \tan\theta) = 1 + \sec\theta \operatorname{cosec}\theta$ [4]

18. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45°. Determine the height of the tower. (Take $\sqrt{3} = 1.732$) [4]

19. Two poles of equal heights are standing opposite each other on either side of a road, which is 80 m wide. From a point between them on the road, the angles of elevation of the tops of the poles are 60° and 30° respectively. Find the height of the poles and the distances of the point from the poles. [4]

— End of Question Paper —